

Excel Sampling Tool — Easy Step-by-Step Explanation

This guide explains the Sampling window in your screenshot.

1. What is Sampling?

Sampling means selecting a smaller group of observations from a larger dataset.

Example: From 50 observations, select 10 observations as a sample.

2. Input Range — \$A\$1:\$A\$51

Input Range tells Excel where the original data is located.

Input Range = \$A\$1:\$A\$51

Excel uses cells A1 through A51. If A1 is a heading, A2:A51 contains 50 observations.

3. Labels ✓

Labels tells Excel that the first cell is a heading, not an observation.

Example: A1 = Sales; A2 = 105; A3 = 112.

4. Sampling Method

Excel provides Periodic and Random sampling.

Method	Meaning
Periodic	Select observations at a fixed interval.
Random	Select observations randomly.

5. Periodic Sampling

Periodic means taking observations at a regular fixed interval.

Example: Period = 5 means sample at a fixed interval of 5 observations.

Easy meaning: Period = how frequently you want to sample.

6. Period Box

The Period box is used when Periodic is selected.

Period = 2 → every 2nd observation. Period = 5 → every 5th observation. Period = 10 → every 10th observation.

7. Random Sampling

Random sampling selects observations without following a fixed interval.

Example: Number of Samples = 10 means Excel selects 10 observations randomly.

8. Number of Samples

This box is used when Random is selected. It means how many observations you want.

9. Output Range — \$D\$1

Output Range tells Excel where to place the result.

Output Range = \$D\$1 means the result starts at cell D1.

10. New Worksheet Ply

Excel creates a new worksheet for the sampling result.

11. New Workbook

Excel creates a completely new workbook containing the sampling result.

12. Your Screenshot — Settings Explained

Option	Your setting	Meaning
Input Range	\$A\$1:\$A\$51	Use data from A1 to A51.
Labels	✓	First cell is a heading.
Sampling Method	Periodic	Use a fixed interval.
Period	Empty	Enter an interval, e.g. 5.
Output Range	\$D\$1	Start output at D1.

13. How to Perform Periodic Sampling

Step 1 → Data → Data Analysis → Sampling.

Step 2 → Input Range: \$A\$1:\$A\$51.

Step 3 → Tick Labels if A1 is a heading.

Step 4 → Select Periodic.

Step 5 → Enter a Period, for example 5.

Step 6 → Output Range: \$D\$1.

Step 7 → Click OK.

14. How to Perform Random Sampling

Step 1 → Open Data → Data Analysis → Sampling.

Step 2 → Input Range: \$A\$1:\$A\$51.

Step 3 → Tick Labels if A1 is a heading.

Step 4 → Select Random.

Step 5 → Enter Number of Samples, for example 10.

Step 6 → Output Range: \$D\$1.

Step 7 → Click OK.

15. Easy Memory Trick

Term	Remember it as
Input Range	Where is my original data?
Labels	Is the first cell a heading?
Periodic	Every fixed interval
Period	How often?
Random	Pick randomly
Number of Samples	How many?
Output Range	Where should the answer go?

One-line summary: Periodic = fixed interval. Random = random selection. Period = how often. Number of Samples = how many. Output Range = where the result goes.